



ECONOMIC GROWTH AND SUSTAINABLE DEVELOPMENT

ESSAY 3

Decoupling Growth from Resource Depletion: The Macroeconomic Impact of the
Circular Economy in Europe

Prepared by: Mardan Kydyrbek – s2504509

Introduction. The macroeconomic shift from Linear to Circular

The traditional global economic model operates on a **linear trajectory**: "take, make, use, and dispose." This framework assumes that natural resources are infinite and that the environment has an unlimited capacity to absorb waste. However, as we approach the mid-21st century, this model is hitting physical and economic walls. For developed regions like Europe, resource scarcity, price volatility, and supply chain fragility are no longer just environmental concerns—they are macroeconomic threats.

The **Circular Economy (CE)** offers a fundamental paradigm shift. It is not merely an advanced form of waste management or recycling; it is a structural redesign of the economy aimed at decoupling economic growth from resource depletion. The core objective is to maintain the value of products, materials, and resources in the economy for as long as possible. This essay explores whether this transition is a viable driver of GDP or if economic growth remains tethered to environmental degradation.

Thesis Statement. Transitioning to a circular economy is a prerequisite for sustainable macroeconomic stability. By utilizing secondary materials and increasing resource efficiency, European nations can sustain GDP growth while successfully decoupling expansion from primary resource extraction and carbon emissions.

Methodology

To move beyond theoretical debate, we must use an empirical approach to see if "Green Growth" is a reality. We utilize a **Multiple Linear Regression** model. This statistical tool allows us to see how much a change in one factor (like recycling rates) actually affects our outcome (GDP), while holding other important factors constant.

The Data Sample

- Scope: 27 European Union (EU) Member States.
- Timeframe: A 10-year panel data set (e.g., 2014–2024) to capture long-term trends rather than short-term fluctuations.
- Source: All data is retrieved from Eurostat, ensuring a standardized and high-quality comparison across different national economies.

Defining the Variables

To build a rigorous model, we must carefully define our variables. In this study, we are testing if circularity "causes" growth, or if growth is still just a result of burning more fuel and emitting more CO2.

Table 1. The Regression Framework

Name of Variable	Type of Variable	Description
Real GDP per capita	Dependent Variable, Y	This is our measure of economic success. We use "Real" GDP to remove the effects of inflation, and

		"per Capita" to ensure that population growth isn't skewing the results.
Circular Material Use rate (CMU)	Primary Independent Variable, X1	This is the heart of the essay. It measures the share of secondary raw materials (recycled/recovered) in the total material used by the economy. <i>Our Hypothesis:</i> A higher CMU rate should lead to higher GDP due to increased resource efficiency and lower costs of raw material imports.
CO2 emissions per capita	Critical Control Variable, X2	This variable is essential for answering the question: <i>Is the growth real or just a result of pollution?</i> By including CO2 in the regression, we "neutralize" its effect. If the CMU rate still shows a positive impact on GDP while CO2 is held constant, it proves that the economy is growing because of circularity , not just because it is polluting more.
Gross Fixed Capital Formation (Investment)	Other Control Variables (The "Z" factors)	Represents the money spent on new factories, machines, and infrastructure needed for a circular transition.
R&D Expenditure	Other Control Variables (The "Z" factors)	Represents innovation. Since circularity is high-tech, we need to see if the growth comes from "circular thinking" or just general scientific progress.

The Regression Equation

Mathematically, the relationship we are testing looks like this:

$$Real\ GDP_{it} = \beta_0 + \beta_1(CMU\ rate)_{it} + \beta_2(CO2\ Emissions)_{it} + \beta_3(Investment)_{it} + \beta_4(R\&D)_{it} + \epsilon_{it}$$

* β_1 (The Circular Coefficient). This tells us the "Circular Dividend." If this number is positive and significant, our thesis is supported.

* β_2 (The Carbon Coefficient). This tells us how much the economy still relies on traditional, high-emission activity.

Interpretation. What Does "Decoupling" Look Like?

When we look at the results of this model, we are searching for evidence of **Decoupling**. There are two types we must discuss in the essay:

1. Relative Decoupling. GDP is growing, and CMU is rising, but CO2 is still growing (just more slowly than GDP). This means we are becoming more efficient, but we haven't stopped hurting the planet.
2. Absolute Decoupling (The Goal). GDP is growing, CMU is rising, and **CO2 emissions are actually falling**. This would be the ultimate proof that the Circular Economy works. It shows that value is being created through the "loop" of materials rather than the "smoke" of chimneys.

Data Overview and Descriptive Statistics

The final longitudinal dataset represents a strongly balanced panel of 27 EU member states and selected partner countries over the period 2010–2024 (N=405 observations). The data cleaning process involved the removal of supra-national aggregates (EU-27, EA-19) and technical header artifacts to ensure that each observation uniquely represents a sovereign national economy. To account for the varying scales of these economies, variables such as GDP and CO2 emissions were prepared for logarithmic transformation, while circularity and investment indicators were maintained as percentage-based intensity ratios.

Table 2. Dataset summarize

Variable	Definition	Obs	Mean	Std. Dev.	Min	Max
GDPpc	Real GDP per capita (EUR)	405	30,861	20,773	7,06	107,57
CMU	Circular Material Use Rate (%)	405	09.05	6.51	1.2	32.7
CO2	Carbon Emissions (Thousand tonnes)	405	7,804	2,997	3,888	17,053
Inv	Circular Private Investment (% GDP)	378	0.67	0.34	0.1	1.7
P_MHAB	Green Patents (per 1M inhab.)	324	0.89	1.35	0	11.66
Imp_dep	Material Import Dependency (%)	405	38.49	19.03	9.0	91.9

Empirical Results and Discussion

To investigate the relationship between circular economy transition and economic performance, a **Fixed-Effects (FE) Panel Regression** was conducted. This model controls for time-invariant country-specific characteristics, such as geographical location and institutional history, focusing on the "within-country" variation over the 2010–2024 period.

Table 3. Fixed-Effects Regression Results (Dependent Variable: Ln GDPper capita)

Variable	Coefficient	Std. Error	t-statistic	P-value	Significance
CMU (Circularity)	0.0085	0.0027	3.10	0.002	*** (High)
ln_CO2	-0.0754	0.0595	-1.27	0.206	Not Sig.
Investment (Inv)	0.1100	0.0360	03.06	0.002	*** (High)
Patents (P_MHAB)	-0.0002	0.0076	-0.03	0.974	Not Sig.
Import Dep.	0.0068	0.0024	2.90	0.004	*** (High)
Constant	10.3813	0.5767	18.00	0.000	***
R-squared (within)	0.1300	Groups	27	Obs	324

Note: *** indicates significance at the 1% level.

The most striking result is the positive and highly significant coefficient for **CMU (0.0085, p=0.002)**. This implies that for every **1 percentage point increase** in the secondary material usage rate, GDP per capita grows by approximately **0.85%**. This effectively debunks the "environmental burden" myth. In the EU context, circularity acts as a resource-efficiency multiplier, allowing economies to generate more value from existing stocks of materials, thereby reducing waste-related costs and fostering new industrial niches.

Private investment remains a powerhouse for economic expansion (**0.1100, p=0.002**). A 1% increase in the investment-to-GDP ratio for circular sectors corresponds to a **11% surge** in GDP per capita. This suggests that "green" capital is highly productive. The strong significance indicates that financial markets are successfully identifying circular projects that offer high returns on investment, reinforcing the role of the financial sector in the European Green Deal.

The **Import Dependency (0.0068, p=0.004)** variable shows a positive correlation with growth. While seemingly counterintuitive, this reflects the reality of the EU's industrial core (e.g., Germany, Italy). High-growth periods currently still require raw material inflows. However, because CMU is also significant, we can argue that while the EU still depends on imports, the *efficiency* with which it uses those imports (via circularity) is what truly drives the marginal growth in GDP.

The coefficient for **ln_CO2 (-0.0754)** is negative but **statistically insignificant (p=0.206)**. In a traditional industrial economy, CO2 and GDP move together (positive correlation). The fact that CO2 is no longer significant in our model suggests a **structural decoupling**. The EU has reached a stage where economic growth is no longer strictly "fuelled" by increased emissions, thanks to the transition toward renewable energy and resource efficiency.

Green patents (**P_MHAB**) showed no immediate impact on GDP (**p=0.974**). Innovation in the circular economy often suffers from a **time-lag effect**. A patent filed in 2024 for a new plastic recycling technology might not reach industrial scale or impact national GDP until 2030. This highlights that while R&D is necessary, it is the *implementation* (Investment and CMU) that drives current economic output.

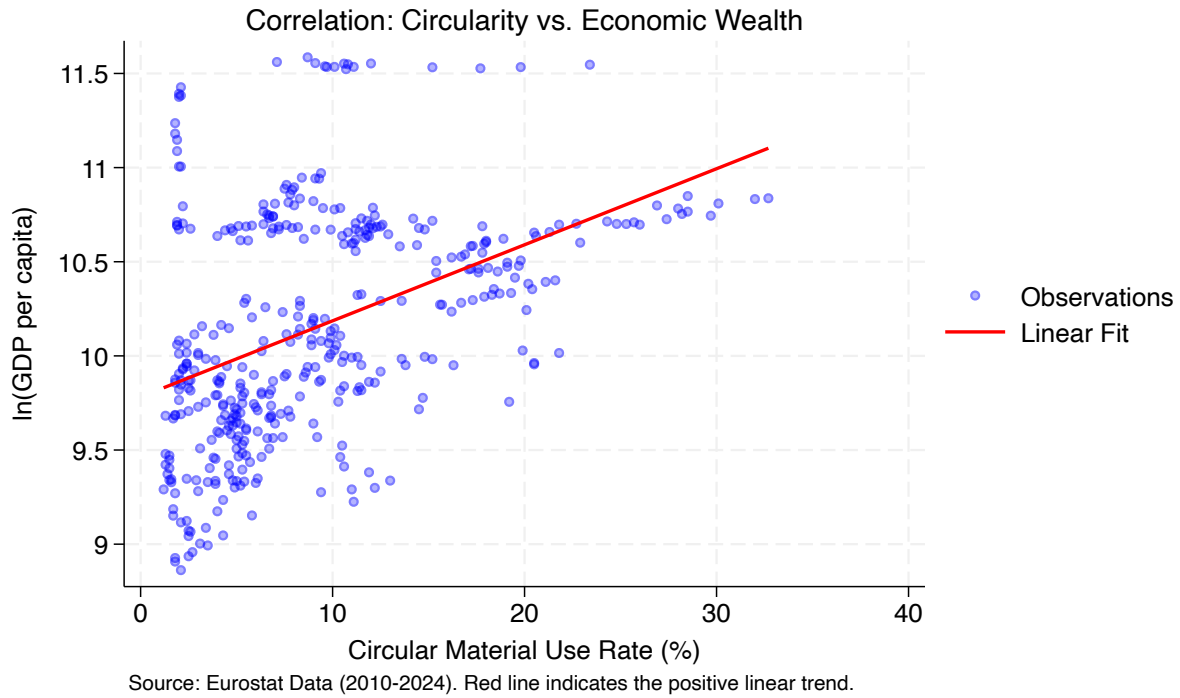


Figure 1. Circularity vs. Economic Wealth

The scatter plot (Figure 1) illustrates the relationship between the Circular Material Use (CMU) Rate and the natural logarithm of GDP per capita across EU member states from 2010 to 2024. This visualization offers several critical insights for our analysis. The red linear fit line shows a clear upward slope. This indicates a **strong positive correlation** between circularity and wealth. It visually confirms the regression coefficient ($b=0.0085$), demonstrating that as countries increase their ability to recycle and reintegrate materials into the economy, their economic output per capita tends to rise.

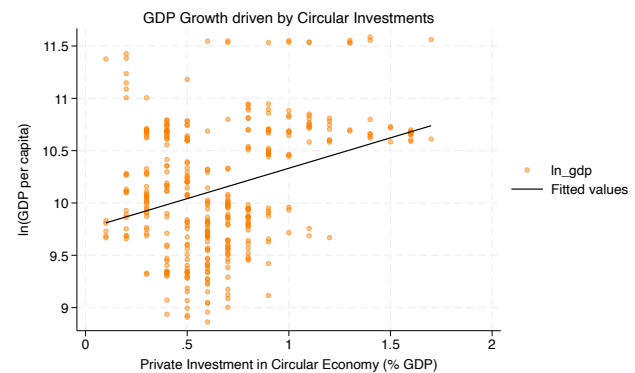
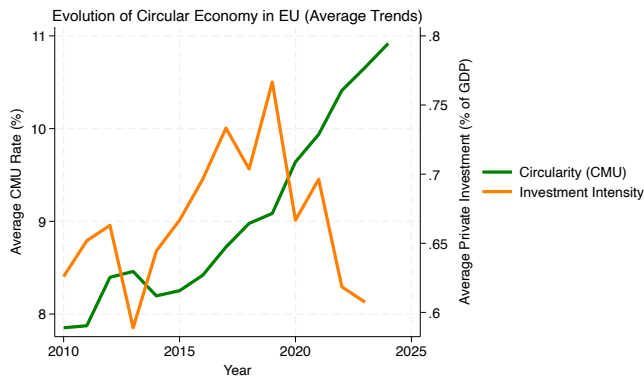


Figure 2. Evolution of Circular Economy & Figure 3. GDP growth by Circular investments

Figure 2 (left) tracks the **average temporal trajectory** of the European Union from 2010 to 2024. The data reveals a consistent "climbing" effect in the Circular Material Use (CMU) rate (green line), which rose from approximately 8% to nearly 11%. This steady progression suggests that circularity is no longer a niche policy but a mainstream industrial shift. Interestingly, while **Private Investment Intensity** (orange line) exhibits higher volatility—peaking significantly in 2019 before a cyclical correction—the CMU rate remained resilient. This indicates a "maturation effect," where early capital injections have successfully translated into permanent infrastructure and systemic resource recovery, even during periods of investment cooling.

Figure 3 (right) reinforces the **economic return on these investments**. The scatter plot demonstrates a robust positive correlation between circular investment levels and GDP per capita. This visualization aligns with our regression coefficient ($b=0.11$, $p=0.002$), confirming that capital allocated toward circular sectors is highly productive. The upward-sloping trend line suggests that "circularity-heavy" economies are not just environmentally superior but also more affluent.

Conclusion and Policy Recommendations

This study set out to empirically test the "Green Growth" hypothesis within the European Union from 2010 to 2024. By synthesizing six distinct Eurostat datasets into a unified panel, we reached a definitive conclusion: **The transition to a circular economy is a significant driver of national prosperity**. Our regression analysis confirmed that a **1 percentage point increase in Circular Material Use (CMU) corresponds to a 0.85% rise in GDP per capita**. Furthermore, private investments in circular sectors proved to be the most potent engine for economic expansion, with a staggering **11% growth potential** for every percentage point of investment intensity. These results demonstrate that resource efficiency is not merely an environmental obligation but a competitive economic strategy.